

Intraday to Three-Day Long Straddle and Strangle Strategies for BTC and ETH Options

Executive summary

Long straddles and strangles (buying both calls and puts) are **long volatility / long gamma** positions: they can profit from large directional moves in either direction, typically at the cost of **time decay (theta)** and exposure to **implied volatility (IV) changes (vega)**. ¹ In most liquid options markets, a naive “buy straddles all the time” approach tends to underperform because option buyers often pay a **volatility (variance) risk premium** and transaction costs; equity-index evidence documents that delta-neutral straddles can have negative average returns. ² Crypto options research and market structure also support the idea that **liquidity and hedging frictions matter** and that risk premia can favor systematic option sellers in the long run, implying that “always long straddles” is unlikely to be robust without timing/filters. ³

A credible path to making **intraday to 2–3 day long straddles/strangles** more viable is to treat them as **conditional bets on volatility expansion**, entered only when (a) the underlying is in a **compression regime** and (b) price/volume structure suggests an elevated probability of a near-term breakout or catalyst-driven jump, while (c) IV is **not already pricing that break too richly**. ⁴

This report provides three strategy families, each with **precise, testable entry/exit rules**: - **Compression-Breakout Long Straddle (CBS)**: buy a very short-dated **ATM straddle** after a volatility compression setup and an objective breakout trigger (price + volume + value-area break). ⁵ - **Value-Area Expansion Long Strangle (VAX)**: buy a slightly **OTM strangle** (defined by deltas) when price escapes a volume-profile “value area” with volume confirmation; cheaper premium but needs a larger move. ⁶ - **Event Window Long Straddle (EWS)**: systematically target **known event windows** (macro releases, large options expiries, etc.) with strict IV and stop rules to avoid paying for already-priced events; this is the most sensitive to IV crush and execution. ⁷

Because you requested a “profitable” strategy: **profitability cannot be asserted without a full backtest** on your exact venue, fee tier, and execution assumptions. This report therefore (1) grounds the strategy design in primary venue rules and empirical literature, (2) specifies a rigorous **backtest blueprint** for 2021–2026 (or the longest available high-quality period), and (3) provides **pseudocode + implementation checklists** to produce auditable results. ⁸

Venue and instrument mechanics that matter for one to three-day options

Contract style, expiry timing, and settlement conventions

On major crypto options venues, options are generally **European style** (exercisable only at expiry) and can be traded/closed prior to expiry. Deribit's educational materials explicitly describe European-style crypto options and that traders can close positions before expiry. ⁹ On Deribit specifically, expiry and settlement conventions (including **08:00 UTC** expiries and TWAP-based settlement windows) are operationally important for very short-dated trading because your last "clean exit" may be hours before your local day ends. ¹⁰

For regulated U.S. venues, **CME** introduced and lists weekly options structures that can match 1–3 day horizons. An official CME Special Executive Report details Monday–Friday weekly options on Bitcoin and Ether futures, with **European exercise** and trading termination at **4:00 p.m. London time** on the named day, and notes that expiring options **exercise into the nearest underlying futures contract**. ¹¹ This "exercise into futures" feature is a major operational risk if you unintentionally hold through expiry.

Binance's options documentation (updated January 2026) states that **options buyers have no margin requirement** beyond paying premium, and that downside risk is limited to premium for buyers. ¹² (Availability, contract specs, and jurisdictional access can change; Binance explicitly flags that the FAQ may be outdated and points users to contract specs and rules effective January 2026. ¹²)

Fees, mark prices, and why execution quality dominates short holding periods

Deribit's fee schedule (updated February 2026) specifies that BTC and ETH options pay a fee equal to **0.03% of the underlying per option contract**, capped at **12.5% of the option price**, and also specifies a separate formula for **USDC-settled linear options** (0.03% of index price, capped at 12.5% of option price). ¹³ For short-dated strategies, this cap matters: cheap near-expiry options can hit the cap quickly, making round trips costlier in percentage terms.

Deribit also uses **mark prices** as reference prices for unrealized P&L, margin, and liquidations, and documents a mark-price methodology intended to be robust when order books are thin or distorted. ¹⁴ Your backtest must decide whether to mark to mid, mark price, or executable quotes; the wrong choice can materially bias results.

Finally, research on crypto option markets shows that **illiquidity/hedging costs are economically meaningful** and affect delta-hedged returns and spreads using transaction-level data from Deribit (Jan 2020–Jul 2024). ¹⁵ Even if you are not delta-hedging, the same microstructure frictions drive the cost of entering/exiting multi-leg positions quickly.

Margin and capital usage for long options

For **long options on Deribit in a Standard Margin account**, the premium is paid in full at purchase and (per Deribit documentation) **there are no further margin requirements** beyond the premium paid; this is stated for both USDC linear options and inverse options. ¹⁶ Deribit also cautions that if you use **portfolio**

margin, you may not need to post the full premium (creating leverage) and **liquidation becomes possible**—a key consideration for “long-premium-only” strategies. ¹⁷

If you enable cross-collateral and run negative balances, Deribit charges **collateral fees** (default 0.05% per day) on the negative equity. ¹⁸ In practice, this means your capital/margin design must also specify collateral management rules, not just trade sizing.

Strategy design with variants and precise entry triggers

Core position variants: straddle vs strangle, ATM vs OTM

A **long straddle** is a call + put at the same strike and expiry; a **long strangle** is a call and put at different strikes (typically OTM). ¹⁹

ATM straddles are generally: - Higher premium, lower breakeven move requirement than an OTM strangle for the same expiry. - Higher **gamma near spot**, so P&L responds more strongly to near-term movement. ²⁰

OTM strangles are generally: - Lower premium (often substantially), but require a larger move to overcome premium and decay. - Lower gamma near spot and more convex “tail” exposure. ²⁰

For a 1–3 day strategy, ATM straddles are usually the “default” if your edge is **breakout magnitude** (you expect a move soon, but not necessarily huge). OTM strangles are more appropriate if your edge is **tail events** (you are selectively targeting jump risk or sharp momentum), and you want cheaper carry. ²¹

A robust entry framework: “setup + trigger + option filter + liquidity filter”

The most defensible short-dated long-premium framework is a **two-stage entry**:

Stage A — Setup: volatility compression / balance - Bollinger BandWidth percentile low: BandWidth measures the percent width between upper and lower Bollinger Bands and is used as a volatility compression proxy. ²²

Suggested filter: $\text{BBWidth_percentile_90d} \leq 20\%$ (i.e., today is within the lowest 20% of BandWidth readings over the past 90 days on your chosen timeframe, e.g., 15m or 1h). - **Realized volatility / ATR compression**: ATR is a volatility measure originally introduced by Wilder and is widely used to quantify “range expansion / contraction.” ²³

Suggested filter: $\text{ATR\%_percentile_90d} \leq 25\%$, where $\text{ATR\%} = \text{ATR} / \text{Spot}$. - **Volume profile balance** (optional but powerful in crypto): define a rolling “session” (e.g., last 24h or a fixed daily cut) and compute Value Area and Point of Control (POC). TradingView defines POC as the price level with highest traded volume and Value Area as the price range for a specified percent (commonly 70%) of volume. ²⁴

Suggested filter: spot spends at least X hours inside the Value Area (e.g., ≥ 8 of the last 12 hours), and the Value Area width is itself low (e.g., VA width percentile $\leq 30\%$).

Stage B — Trigger: objective breakout / impulse - Value Area escape: enter only if price **closes** outside Value Area High (VAH) or below Value Area Low (VAL) and stays outside for at least two candles on your execution timeframe (e.g., 2x5m closes). (VAH/VAL/POC definitions: ²⁴) - **Volume confirmation**: breakout

candle volume $\geq 1.5 \times$ the 20-candle average volume, or a comparable volume impulse rule (volume profile is explicitly volume-based). ²⁵ - **Optional momentum confirmation**: “squeeze release” style logic can be implemented via BandWidth rising after compression (BandWidth turning up from low percentile), consistent with the “compression then expansion” interpretation of BandWidth. ²⁶

Stage C — Option pricing filter: avoid paying “already expensive” IV Since implied volatility reflects market expectations derived from option prices, entering long premium when IV is already elevated often worsens expectancy. ²⁷ Your filters should include at least one of:

- **IV rank / percentile** (computed on your dataset):
- $\text{IV_percentile_180d} \leq 50\%$ for the targeted expiry bucket (e.g., 1–3 DTE ATM IV).
- Stricter variants: $\leq 30\%$ (more selective, fewer trades).
- **Term structure sanity check**: avoid entering long premium when front-end IV is extremely elevated relative to the next expiry (a common “event priced” signature), unless your strategy is explicitly event-driven. Deribit analytics reports frequently discuss term structure shifts and short-tenor IV moves, illustrating the importance of tenor-specific IV. ²⁸

A simple rule: require $\text{IV}(1\text{--}3\text{DTE ATM}) \leq \text{IV}(7\text{DTE ATM}) + 5 \text{ vol points}$ for breakout entries.

Stage D — Greeks and microstructure filter - Delta neutrality: ensure the combined position’s absolute delta is small at entry (e.g., $|\Delta_{\text{total}}| \leq 0.05$). Delta is a core Greek measuring sensitivity to underlying price. ²⁰ - **Gamma sufficiency**: for positions meant to profit from quick movement, require Gamma_total above a minimum threshold (threshold must be calibrated in backtest; gamma is highest near ATM and short expiries). ²⁰ - **Spread filter**: reject trades where the sum of bid-ask spreads implies an excessive round-trip cost. Illiquidity is empirically important in crypto option returns and spreads. ¹⁵

Example rule: $(\text{Ask-Bid})/\text{Mid} \leq 4\%$ for each leg, and combined “two-leg” spread cost $\leq 6\text{--}8\%$ of total premium.

Candidate rulesets table (parameters and logic)

The following rulesets are designed to be **backtestable** without discretionary judgment. Numeric thresholds are starting points; you should expect to tune them via walk-forward training and holdout testing.

Ruleset	Structure	DTE window	Strike selection	Setup filters	Trigger	Option filters	Best use-case
CBS	Long ATM straddle	1–2 days	Closest strike to spot (ATM)	BBWidth percentile $\leq 20\%$ (90d); ATR% percentile $\leq 25\%$ (90d) ²⁹	2× closes outside VAH/VAL + volume impulse ²⁵	IV percentile $\leq 50\%$ (180d)	Range breakouts after compression

Ruleset	Structure	DTE window	Strike selection	Setup filters	Trigger	Option filters	Best use-case
VAX	Long OTM strangle	2-3 days	Put $\approx -0.25\Delta$, Call $\approx +0.25\Delta$ (or nearest) 20	Value Area width percentile $\leq 30\%$ (lookback 60-90d) 25	Close outside Value Area + retest failure (no return to VA in next N candles)	IV percentile $\leq 40\%$ + skew not extreme	"Fast trend day" / larger moves
EWS	Long ATM straddle	1-3 days	ATM	No compression needed	Enter T-60 to T-15 min before scheduled event; exit post-event	Must have IV not already "spiked" vs prior 24h	Known catalysts; strict risk controls
CBS-Lite	Long ATM strangle (tight)	1-2 days	Put/Call $\approx \pm 0.35\Delta$	BBWidth percentile $\leq 15\%$	Breakout close beyond BB + volume impulse 30	IV percentile $\leq 50\%$	Cheaper than straddle, still "near"

Volume profile definitions used above (POC and Value Area) are taken from TradingView and CQG documentation describing how value area is typically computed from POC outward to reach ~70% of volume/TPOs. 24

Flowchart for automated entry decision

flowchart TD

```

A[Start: every new 5m/15m bar] --> B{DTE in range?}
B -- No --> Z[No trade]
B -- Yes --> C{Setup: compression?}
C -- No --> Z
C -- Yes --> D{Trigger: VA breakout + volume impulse?}
D -- No --> Z
D -- Yes --> E{Option filter: IV percentile ok? spreads ok? delta ~ 0?}
E -- No --> Z

```

```
E -- Yes --> F[Enter: buy call + buy put (straddle/strangle)]  
F --> G[Set exits: PT/SL/time/IV]
```

Exit rules, stop logic, and risk management

P&L mechanics you must respect in short-dated long premium

A practical option P&L decomposition uses Greek sensitivities: delta, gamma, theta, and vega describe how option value changes with underlying price, curvature, time decay, and volatility. ³¹ More advanced frameworks explicitly link short-term option investment outcomes to daily P&L attribution. ³²

For your use case (close before expiry), you want exits that convert the position from “needs movement soon” into “harvest profit quickly and avoid giving it back.”

Canonical exit framework (suitable for automation)

Below are concrete exit rules you can implement exactly; choose one “primary” rule and keep others as safety rails.

Profit-taking rules - PT1 (premium multiple): close both legs when mid/mark value of the combined position reaches

$\text{Value_now} \geq 1.30 \times \text{Premium_paid}$ (30% gross profit).

Alternate PT ladder: 20% / 35% / 50% depending on regime. - **PT2 (move vs breakeven):** compute breakeven move in dollars:

$\text{BE\$} = \text{Premium_paid} \times \text{Spot}$ (if premium expressed as fraction of spot; venue-specific).

Close when $|\text{Spot} - \text{Spot_entry}| \geq 0.8 \times \text{BE\$}$ AND the position is net profitable after costs.

(This is designed to take profits before true expiry breakeven because gamma profits are front-loaded while theta accelerates.)

Stop-loss rules - SL1 (premium drawdown): close if

$\text{Value_now} \leq 0.60 \times \text{Premium_paid}$ (40% loss) OR if **time stop** hits first. - **SL2 (time decay stop):** close if

$\text{Time_in_trade} \geq 18 \text{ hours}$ for 1–2 DTE trades (or ≥ 24 hours for 2–3 DTE) AND the underlying has not moved at least $0.35 \times \text{BE\$}$.

Rationale: short-dated long premium is mostly a race between movement and theta. ³³

IV reversion / crush rules - IV-stop: if the ATM IV for your expiry drops by more than a set threshold from entry (example: $\Delta \text{IV} \leq -8 \text{ vol points}$) and spot has not already moved $> 0.5 \times \text{BE\$}$, exit.

Implied volatility is directly derived from option prices and can shift rapidly; realized vs implied differences matter for strategy expectancy. ³⁴

Greeks-based risk cap (optional but clean) - Delta cap: if $|\Delta_{\text{total}}| \geq 0.35$ and you are not delta-hedging, either (a) close the trade, or (b) convert to a single-leg position by selling the far-OTM “losing” leg (this is a strategy change; test separately). Delta/gamma explain the increasing directional exposure after a move. ²⁰

Hard operational exit - Close all positions no later than a fixed time before expiry. On Deribit, IV orders are canceled **15 minutes before option expiry**, which can affect automated execution if you rely on IV orders.

³⁵ On CME weekly options, expiring options exercise into futures, so you may prefer closing well before the termination time to avoid unintended futures positions. ¹¹

Position sizing and portfolio limits

Because a long straddle/strangle has **limited loss equal to premium**, sizing is conceptually simpler than short options—but **portfolio ruin can still happen** via serial small losses.

A conservative sizing regime for a short-dated long-premium program: - **Max loss per trade**: 0.50%–1.00% of portfolio NAV (premium + fees).

For highly volatile conditions or thin liquidity, cap at 0.25%–0.50%. (Crypto options are high vol; hedging and liquidity frictions are documented concerns. ³⁶) - **Max concurrent exposure**: at most 2–4 open long-premium positions across BTC/ETH combined, with a tighter cap if positions share the same directional driver (BTC and ETH are often correlated in stress).

- **Loss limits**: stop trading for the day after a fixed drawdown (e.g., $-2R$ or $-3R$), and for the week after $-6R$ (where R = max loss per trade). - **Capital/margin**:

- On Deribit Standard Margin, long options require paying premium in full and have no further margin requirement (so the “capital at risk” is premium). ¹⁶

- If using portfolio margin, explicitly cap leverage because long options can be liquidated when the system does not require full premium posted. ¹⁷

- If using cross collateral, avoid negative settlement-currency equity to eliminate collateral fees. ¹⁸

Transaction cost and slippage assumptions (explicit)

You asked for explicit assumptions; below is a baseline you can refine after measuring venue-specific spreads.

Deribit fee model (primary, explicit) - BTC options trading fee: $\text{MIN}(0.0003 \text{ BTC}, 0.125 \times \text{OptionPrice}) \times \text{Amount}$ ¹³

- ETH options trading fee: $\text{MIN}(0.0003 \text{ ETH}, 0.125 \times \text{OptionPrice}) \times \text{Amount}$ ¹³

- USDC linear BTC/ETH options: $\text{MIN}(0.0003 \times \text{IndexPrice}, 0.125 \times \text{OptionPrice}) \times \text{Contracts}$

¹³

Execution slippage assumptions (for planning; validate empirically) - Base case: fill at **mid + 25% of spread** on entry and **mid – 25% of spread** on exit per leg. - Stress case (fast breakout): fill at **ask on entry** and **bid on exit** (worst-case “crossing the spread”). - Add **legging risk** if not executed as a combo: simultaneous execution is preferred to reduce directional drift between legs (especially around breakouts).

CME and Binance high-level constraints - CME weekly options have fixed tick sizes and exchange fees specified in CME notices; your all-in costs will depend on broker/clearing. ¹¹

- Binance states options buyers have no margin requirement beyond premium, but you must source exact fee schedules and contract specs from Binance’s linked documents for precise modeling. ¹²

Sensitivity to IV level, move magnitude, and time to expiry

Breakeven move: a central design variable

A common “expected move” heuristic is: - **Expected move (1σ)** $\approx$ $\text{Spot} \times \text{IV} \times \text{sqrt}(T)$ (with T in years). ³⁷

Practitioners also use the **ATM straddle price** as a “quick estimate” of expected move (straddle premium approximates the market’s priced move). ³⁸

A more model-based rule-of-thumb for an ATM straddle under idealized assumptions uses the relationship between standard deviation and absolute deviation (the $\sqrt{2/\pi}$ factor), referenced in the literature around implied volatility approximations and the Brenner–Subrahmanyam framework. ³⁹

Practical breakeven table (illustrative, model-based)

Let: - **IV** = annualized implied volatility (decimal, e.g., 0.60) - **DTE** = days to expiry - **T** = $\text{DTE}/365$ (crypto trades 24/7; BVIN methodology explicitly uses 365 days rather than 252 for annualization). ⁴⁰ - Approximate 1σ move $\approx$ $\text{IV} \times \text{sqrt}(T)$ - Approximate “absolute move scale” $\approx$ $0.80 \times \text{IV} \times \text{sqrt}(T)$ ($\sqrt{2/\pi} \approx 0.80$; see implied-vol approximation references). ⁴¹

IV (annualized)	1 DTE: $0.80 \times \text{IV} \times \sqrt{1/365}$	2 DTE	3 DTE
40%	~1.7%	~2.4%	~2.9%
60%	~2.5%	~3.6%	~4.4%
80%	~3.4%	~4.8%	~5.9%
100%	~4.2%	~6.0%	~7.3%

Interpretation: if 2 DTE ATM IV is 80%, a rough “move scale” is near ~4.8% over that horizon; if you pay a straddle premium larger than that (after costs), your required move must be correspondingly larger, or you need IV expansion to offset. ⁴²

Mermaid chart: breakeven move vs IV for a two-day horizon (illustrative)

```
xychart-beta
  title "Move scale ( $0.80 \times \text{IV} \times \text{sqrt}(2/365)$ ) vs IV"
  x-axis "IV (annualized %)" [40, 60, 80, 100]
  y-axis "Move scale (%)" 0 --> 8
  line [2.4, 3.6, 4.8, 6.0]
```

Why IV sensitivity dominates event trades

Your position wins via: 1) **Spot move** big enough and fast enough (gamma P&L),
2) **IV rising** after you enter (vega P&L),
and loses mainly via **theta** and **IV collapse**. The Greeks summarize these exposures. ³¹

This is why “event straddles” can fail: IV often rises into the event and falls immediately after (IV crush), so you must demand either (a) a move larger than priced, or (b) an entry where IV is *not yet elevated*. The “expected move” framing is widely used to reason about this. ³⁸

Backtest blueprint, data requirements, and implementation scaffolding

Data required (minimum viable vs “research grade”)

Minimum viable backtest (will be biased, but can screen ideas) - Underlying spot or futures candles (1m to 5m): OHLCV. - Option mid prices for the chosen strikes/expiries at your decision frequency (e.g., 1m or 5m). - ATM IV for the chosen expiry bucket to compute IV percentiles and IV stops. - Fee model and a conservative spread/slippage model.

Research-grade backtest (recommended) - Tick or 1-second **trades and quotes** for options (bid/ask), not just prints. - Snapshot or incremental L2 book for underlying (spot/futures/perp) to model breakout slippage. - Full IV surface (strike × expiry) and Greeks at high frequency (or reconstruct using a model such as SVI; crypto options literature explicitly uses SVI to address sparse/stale quotes). ⁴³ - Funding rates if you delta-hedge with perps (Deribit notes funding is a transfer between longs/shorts and Deribit does not charge fees on funding). ⁴⁴ - Margin/collateral state if using cross-collateral (collateral fees exist for negative balances). ¹⁸

Exchanges/data sources to prioritize and how to obtain them

Deribit (primary for crypto options liquidity) - Academic and industry sources note Deribit’s dominance in crypto options liquidity; recent research cites Deribit as the largest venue and analyzes Deribit trade-level data. ⁴⁵

- Deribit official API provides: - OHLCV: `public/get_tradingview_chart_data` ⁴⁶
- Trades: `public/get_last_trades_by_currency` and by instrument/time ⁴⁷
- Historical access via a `historical` parameter for trade/order retrieval (Deribit documentation and support explain retention and historical retrieval). ⁴⁸
- Historical volatility: `public/get_historical_volatility` ⁴⁹
- DVOL candles: `public/get_volatility_index_data` ⁵⁰
- Order book snapshots: `public/get_order_book` ⁵¹
- Mark price history (limited subset of options): `public/get_mark_price_history` ⁵²

High-frequency historical quotes / L2 books - Deribit partnered with Tardis.dev ⁵³ to provide high-frequency historical datasets (e.g., tick-by-tick trades, order book updates). ⁵⁴
This is the cleanest path to a research-grade backtest without building your own recorder.

CME - Use CME instruments for a regulated comparison set and for strategies that prefer standardized clearing. CME’s SER-9181 provides contract specs for weekly options on Bitcoin/Ether futures, trading hours, termination time, and European exercise into futures. ¹¹
- Historical intraday options quotes may require CME market data vendors or CME DataMine (not covered by the provided sources; treat as an external procurement step).

Pseudocode for backtesting (event-driven, realistic fills)

```
# PSEUDOCODE (venue-agnostic). No external dependencies assumed.

for t in timeline: # 1m or 5m bars
    update_underlying_bars(t)
    update_volume_profile(t) # compute POC/VAH/VAL for chosen session window
    update_indicators(t)     # BBWidth, ATR%, volume impulse metrics
    update_option_chain(t)   # for target expiries compute mid, bid/ask,
    greeks, IV

    # --- Identify eligible expiries ---
    eligible_expiries = [e for e in expiries if 1 <= DTE(e, t) <= 3]

    for expiry in eligible_expiries:
        # --- Stage A: Setup ---
        if BBWidth_percentile(t) > 0.20:
            continue
        if ATRpct_percentile(t) > 0.25:
            continue
        if value_area_width_percentile(t) > 0.30:
            continue

        # --- Stage B: Trigger ---
        breakout_dir = None
        if close_outside_VA_for_N_bars(t, N=2, side="above"): breakout_dir =
"up"
        if close_outside_VA_for_N_bars(t, N=2, side="below"): breakout_dir =
"down"
        if breakout_dir is None:
            continue
        if not volume_impulse(t, mult=1.5):
            continue

        # --- Stage C: Option filter ---
        atm_iv = get_atm_iv(expiry, t)
        if iv_percentile(expiry_bucket="1-3DTE_ATM", t=t) > 0.50:
            continue

        # Liquidity filter: skip if spreads too wide
        call, put = select_legs(strategy="CBS", expiry=expiry, t=t)
        if rel_spread(call, t) > 0.04 or rel_spread(put, t) > 0.04:
            continue

        # --- Stage D: Greeks filter ---
        if abs(delta(call,t) + delta(put,t)) > 0.05:
            continue
```

```

    # --- Simulate entry execution ---
    entry_price_call = simulate_fill(call, side="buy",
model="mid+0.25spread")
    entry_price_put = simulate_fill(put, side="buy",
model="mid+0.25spread")
    entry_fees = fee_model(call, entry_price_call) + fee_model(put,
entry_price_put)

    open_trade = {
        "t_entry": t,
        "expiry": expiry,
        "call": call, "put": put,
        "premium": entry_price_call + entry_price_put,
        "fees_paid": entry_fees,
        "spot_entry": spot(t),
        "iv_entry": atm_iv,
    }

    # --- Manage trade forward in time ---
    for u in future_times(t, until=min(expiry_cutoff(expiry), t+72h)):
        value_now = mark_to_mid_or_mark(open_trade, u)
        pnl_now = value_now - open_trade["premium"] -
open_trade["fees_paid"] - est_exit_costs(open_trade, u)

        # Profit target
        if value_now >= 1.30 * open_trade["premium"]:
            close_trade(open_trade, u, reason="PT")
            break

        # Stop loss
        if value_now <= 0.60 * open_trade["premium"]:
            close_trade(open_trade, u, reason="SL")
            break

        # IV stop
        if atm_iv_at(u) <= open_trade["iv_entry"] - 8 and abs(spot(u)-
spot(t)) < 0.5 * breakeven_dollars(open_trade):
            close_trade(open_trade, u, reason="IV_stop")
            break

        # Time stop
        if u - t >= 24h and abs(spot(u)-spot(t)) < 0.35 *
breakeven_dollars(open_trade):
            close_trade(open_trade, u, reason="time_stop")
            break

```

Pseudocode for live execution (Deribit-oriented concepts)

Deribit's API supports OHLCV, trades, order books, historical retrieval, and DVOL, and provides operational notes on best practices for market data collection. ⁵⁷

```
# LIVE EXECUTION CHECKLIST PSEUDOCODE (Deribit-like)

connect_ws()
subscribe_to_channels([
    "ticker.*", "book.*", "trades.*", # underlying and options
])

while True:
    t = now_utc()
    update_local_state_from_ws()

    if not risk_ok():
        continue

    signal = compute_signal_from_state() # compression + VA breakout + IV
    filters

    if signal.enter:
        # Use atomic execution if venue supports multi-leg combos;
        # otherwise minimize legging risk by simultaneous orders.
        place_limit_buy(call, price=iv_order_or_limit_price(call))
        place_limit_buy(put, price=iv_order_or_limit_price(put))

        set_exits(trade_id, PT=+30%, SL=-40%, time_stop_hours=24)

    for trade in open_trades:
        if should_exit(trade):
            close_both_legs(trade)
```

Operational notes: - If using Deribit **IV orders**, remember they are canceled **15 minutes** before expiry. ³⁵
- Use WebSocket subscriptions rather than polling for low latency and rate-limit efficiency. ⁵⁸
- Decide whether your automation marks risk to **mark price** (risk engine reference) or executable quotes.

¹⁴

Implementation checklist (practical)

- Define your “clock”: 24/7 crypto requires consistent definitions of “day,” “session,” and rolling windows; BVIN methodology explicitly uses 365-day scaling, which aligns with always-on markets. ⁴⁰
- Pick venue/instrument type: inverse vs USDC linear options on Deribit (linear USDC options are explicitly European, priced and settled in USDC). ⁵⁹

- Encode fee model exactly (Deribit formulas differ for inverse vs USDC linear and include caps). ¹³
- Build trade reconstruction: ensure you can reproduce entry/exit with quotes (not just prints); crypto options can be sparse and stale. ⁶⁰
- Calibrate liquidity filters per expiry bucket (1–3 DTE options can be liquid at ATM but thin at far strikes).
- Run parameter sweeps with **walk-forward validation**; avoid optimizing on the full sample.
- Stress test: widen spreads, apply worst-case fills, and re-run—microstructure is often the difference between “paper edge” and real edge. ⁶¹
- Add operational hard stops: maximum daily loss, circuit breaker on exchange/API degradation, and “no-new-trades” windows around expiry cutoffs.

Concrete trade examples (illustrative templates)

These examples are **structural** (they show how to apply the rules); you should plug in real chain quotes from your venue.

Example A — CBS (ATM straddle, 2 DTE) - Setup: BBWidth percentile (90d, 15m) = 12%; ATR% percentile = 18%. ²⁹

- Volume profile (24h): POC and Value Area computed; price has been inside VA for 10 of last 12 hours. ²⁵
- Trigger: two consecutive 5m closes above VAH; breakout volume = $1.8 \times 20\text{-bar avg.}$ ²⁵
- Option filter: 2DTE ATM IV percentile (180d) = 35% (acceptable). - Entry: buy ATM call + ATM put. Target PT = +30%, SL = -40%, time stop = 24h. - Exit: close when combined value hits $1.30 \times \text{premium}$ or at IV-stop if ATM IV drops by 8 vol points without sufficient spot movement.

Example B — VAX (OTM strangle, 3 DTE, ± 0.25 delta) - Setup: Value Area width percentile (90d) $\leq 30\%$ and BBWidth percentile $\leq 20\%$. ⁶²

- Trigger: close below VAL, then a failed retest (price does not re-enter VA within next $6 \times 5\text{m bars}$). - Entry: buy put at -0.25Δ and call at $+0.25\Delta$ (3 DTE). Delta and gamma concepts guide this selection. ²⁰
- Exit: PT = +40% (because tails required), SL = -35%, time stop = 30h.

Example C — EWS (event straddle) - Enter 30–60 minutes before a scheduled event only if front-end IV is not already elevated relative to its 24h distribution (IV percentile $\leq 40\%$). The “expected move” framing is commonly tied to ATM straddle pricing. ⁶³

- Exit 5–30 minutes after the event if profitable; otherwise apply strict SL/time stop to avoid post-event decay and IV crush.

¹ ⁵ ¹⁹ Multi-leg Options Positions (Part 1 - Straddles and Strangles)

https://insights.deribit.com/education/multi-leg-options-positions-part-1-straddles-and-strangles/?utm_source=chatgpt.com

² Straddles around Earnings Announcements

https://www.pbcsf.tsinghua.edu.cn/_local/A/11/35/B0A1714D795BD9799087BCCC2AD_A01C4283_54641.pdf?e=.pdf&utm_source=chatgpt.com

³ Valuation and hedging of cryptocurrency inverse options by Artur Sepp, Vladimir Lucic :: SSRN

<https://papers.ssrn.com/sol3/Delivery.cfm/4606748.pdf?abstractid=4606748&mirid=1>

- 4 22 26 29 **Bollinger BandWidth - ChartSchool - StockCharts.com**
https://chartschool.stockcharts.com/table-of-contents/technical-indicators-and-overlays/technical-indicators/bollinger-bandwidth?utm_source=chatgpt.com
- 6 24 25 62 **Volume profile indicators: basic concepts**
https://www.tradingview.com/support/solutions/43000502040-volume-profile-indicators-basic-concepts/?utm_source=chatgpt.com
- 7 21 32 56 **Option Profit and Loss Attribution and Pricing: A New ...**
https://engineering.nyu.edu/sites/default/files/2020-09/option-profit-carr-jofi-12894.pdf?utm_source=chatgpt.com
- 8 46 57 **public/get_tradingview_chart_data**
https://docs.deribit.com/api-reference/upcoming/market-data/public-get_tradingview_chart_data?utm_source=chatgpt.com
- 9 **Cryptocurrency Call Options**
https://insights.deribit.com/education/cryptocurrency-call-options/?utm_source=chatgpt.com
- 10 **Bitcoin Futures Contract Specs**
https://www.cmegroup.com/trading/bitcoin-futures.html?utm_source=chatgpt.com
- 11 **cmegroup.com**
<https://www.cmegroup.com/notices/ser/2023/04/SER-9181.pdf>
- 12 **Introduction to Binance Options | Binance Options**
<https://www.binance.com/en/support/faq/detail/374321c9317c473480243365298b8706>
- 13 **Fees – Deribit Support**
<https://support.deribit.com/hc/en-us/articles/25944746248989-Fees>
- 14 **Mark Prices**
https://support.deribit.com/hc/en-us/articles/25944746962973-Mark-Prices?utm_source=chatgpt.com
- 15 36 61 **Illiquidity Premium and Crypto Option Returns**
https://acfr.aut.ac.nz/_data/assets/pdf_file/0006/969378/950002_Atanasova_Illiquidity-Premium-and-Crypto-Option-Returns.pdf
- 16 59 **Linear USDC Options**
https://support.deribit.com/hc/en-us/articles/31424932728093-Linear-USDC-Options?utm_source=chatgpt.com
- 17 **Portfolio Margin**
https://insights.deribit.com/education/portfolio-margin/?utm_source=chatgpt.com
- 18 **Cross collateral specifications**
https://support.deribit.com/hc/en-us/articles/25944777203869-Cross-collateral-specifications?utm_source=chatgpt.com
- 20 31 33 53 **Option Greeks: The 4 Factors to Measure Risk**
https://www.investopedia.com/trading/getting-to-know-the-greeks/?utm_source=chatgpt.com
- 23 **Average True Range (ATR) - ChartSchool - StockCharts.com**
https://chartschool.stockcharts.com/table-of-contents/technical-indicators-and-overlays/technical-indicators/average-true-range-atr-and-average-true-range-percent-atrp?utm_source=chatgpt.com
- 27 **Market Implied Volatility | Product Docs**
https://gitbook-docs.coinmetrics.io/market-data/market-data-overview/volatility/market-implied-volatility?utm_source=chatgpt.com
- 28 **Crypto Derivatives: Analytics Report - Week 48**
https://insights.deribit.com/industry/crypto-derivatives-analytics-report-week-48-2023/?utm_source=chatgpt.com

30 **How to Use Bollinger Bands to Gauge Trends**

https://www.investopedia.com/trading/using-bollinger-bands-to-gauge-trends/?utm_source=chatgpt.com

34 42 **Implied vs Realized Volatility: What is the Difference?**

https://www.tastylive.com/concepts-strategies/implied-vs-realized-volatility?utm_source=chatgpt.com

35 **IV Order - Options**

https://support.deribit.com/hc/en-us/articles/25944809696029-IV-Order-Options?utm_source=chatgpt.com

37 38 **Understanding and calculating the expected move of a stock ...**

https://www.home.saxo/content/articles/equities/understanding-and-calculating-the-expected-move-of-a-stock-etf-index-07072023?utm_source=chatgpt.com

39 **A Simple Formula to Compute the Implied Standard ...**

https://www.jstor.org/stable/4479152?utm_source=chatgpt.com

40 **cryptocompare.com**

<https://www.cryptocompare.com/media/37621858/the-bitcoin-volatility-index-bvin-whitepaper.pdf?v3=>

41 **(PDF) A Simple Formula to Compute the Implied Standard ...**

https://www.researchgate.net/publication/245065192_A_Simple_Formula_to_Compute_the_Implied_Standard_Deviation?utm_source=chatgpt.com

43 60 **arxiv.org**

<https://arxiv.org/pdf/2112.06807>

44 **Funding Specifications**

https://support.deribit.com/hc/en-us/articles/31424939178397-Funding-Specifications?utm_source=chatgpt.com

45 **Price dynamics and volatility jumps in bitcoin options**

https://link.springer.com/article/10.1186/s40854-024-00653-z?utm_source=chatgpt.com

47 **public/get_last_trades_by_currency**

https://docs.deribit.com/api-reference/upcoming/market-data/public-get_last_trades_by_currency?utm_source=chatgpt.com

48 **Accessing Historical Trades and Orders Using API**

https://docs.deribit.com/articles/accessing-historical-trades-orders?utm_source=chatgpt.com

49 **public/get_historical_volatility**

https://docs.deribit.com/api-reference/market-data/public-get_historical_volatility?utm_source=chatgpt.com

50 **public/get_volatility_index_data - Deribit API Documentation**

https://deribit.mintlify.app/api-reference/market-data/public-get_volatility_index_data?utm_source=chatgpt.com

51 **public/get_order_book**

https://docs.deribit.com/api-reference/upcoming/market-data/public-get_order_book?utm_source=chatgpt.com

52 **public/get_mark_price_history**

https://docs.deribit.com/api-reference/upcoming/market-data/public-get_mark_price_history?utm_source=chatgpt.com

54 **Celebrating our Tardis.dev partnership - Get free historical ...**

https://insights.deribit.com/exchange-updates/celebrating-our-tardis-dev-partnership-get-free-historical-data/?utm_source=chatgpt.com

55 **Implied volatility estimation of bitcoin options and the stylized ...**

https://pmc.ncbi.nlm.nih.gov/articles/PMC8418903/?utm_source=chatgpt.com

58 **Deribit Market Data Collection Best Practices**

https://docs.deribit.com/articles/market-data-collection-best-practices?utm_source=chatgpt.com

63 **How to Calculate Expected Move**

https://www.tastylive.com/definitions/calculating-expected-move?utm_source=chatgpt.com